McMaster University | DeGroote School of Business

Pedigree vs. Grit:

Predicting Mutual Fund Manager Fund Performance

Course: MFIN604

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Executive Summary

This report evaluates whether mutual fund manager “pedigree” (undergraduate SAT score and MBA) meaningfully predicts annual excess fund returns and helps AMBTPM choose between Bob Smith and Putney Rockefeller. Using the regression results in Table 1, we find that a low R-squared (4.33%) does not make the model useless, instead, excess returns are inherently volatile, so low explanatory power is typical, yet the regression can still support inference about which characteristics are systematically related to performance.

Here are some key findings: First, fund type matters. Growth and income funds earn significantly lower average excess returns than growth funds. In Table 1, the growth-and-income indicator (GRI) is negative and statistically significant, implying that, holding other manager characteristics constant, growth-and-income funds underperform growth funds on average. Second, undergraduate school quality (SAT) is positively related to excess returns and is statistically significant. A one-tailed test supports the claim that if Bob had attended Princeton rather than Ohio State, his expected excess return would be higher. In contrast, MBA status is not statistically significant. The MBA coefficient is near zero with a very large p-value, so there is no substantial evidence that managers with MBAs perform worse (or better) once fund type, SAT, age, and tenure are held constant. Age shows a modest negative relationship with performance at the approximately 5% level in a one-tailed test, and survivorship bias likely exaggerates the apparent strength of this adverse age effect.

For decision-making, we compare predicted returns for the two candidates. Based on Table 1, Putney’s expected excess return is higher than Bob’s both at their current funds and under an AMBTPM-hire scenario (resetting tenure at the new fund). The expected advantage is economically meaningful (about 1.8 percentage points when hired), although the regression’s considerable residual variation implies substantial uncertainty around any single-year outcome. Taken together, the evidence supports hiring Putney Rockefeller since her higher predicted excess return is driven primarily by the SAT (pedigree) and not by the MBA label itself, and there is no statistical basis to view an MBA as detrimental. We recommend that AMBTPM select Putney, recognizing that returns are noisy. The firm should treat these estimates as a single input and continue to monitor realized performance.

Question 1

(a) Why do you disagree with Jack's comments about the uselessness of the regression due to the low R-squared?

Although the R-squared of 4.33% is low, this does not mean the regression is useless. It indicates that the manager characteristic is not the primary driver of excess fund returns, meaning that other factors drive the performance. Low R-squared values are expected in excess return regressions because they are highly volatile and largely affected by market conditions, investment timing, and other hidden factors. The regression is still helpful for inference. For example, despite the low R-squared, GRI (t=2.156, p=3.151%) and SAT (t=-2.856, p=0.445%) remain significant, indicating that these two characteristics are systematically related to fund performance.

(b) Can you think of a situation in which a useless regression has a high R-squared?

A regression of crime rates on the number of police officers may produce a high R-squared because areas with higher crime tend to be assigned more police officers. However, this relationship is misleading, as police presence responds to crime rather than causes it. Therefore, despite the strong statistical fit, the regression is not useful for drawing meaningful conclusions about the effect of police on crime.

(c) There are techniques to determine the validity of a regression model—in particular, whether the relationship is linear and the error terms display equal variance (homoskedasticity). Does the regression in Table 1 violate either of these two assumptions? Justify your answer.

From Table 1, there is no evidence that the regression violates linearity or homoskedasticity, but using the raw data, linearity can be assessed by examining residuals versus fitted values or residuals versus the explanatory variables. A linear relationship is supported when residuals show no systematic pattern.

The residual plots show residuals randomly scattered around zero, with no clear curvature or systematic change in spread. While the residual plots do not display a pronounced fan- or wedge-shaped pattern, homoskedasticity was formally tested using an auxiliary regression of squared residuals on the explanatory variables (Appendix 1). The F-test from this auxiliary regression yields a p-value of 0.003, leading us to reject the null hypothesis of constant variance. Therefore, based on the regression, there is no evidence that the regression violates the linearity assumption, but it violates the homoskedasticity assumption.

Question 2

(a) Estimate the excess return (RET) of the funds that Bob and Putney currently manage. Assume Princeton's average composite SAT score is 1355 and Ohio State's is 1042. Who is expected to obtain higher returns at their current funds, and by how much?

Bob RET = -2.64216-2.11046(1)+0.005735(1042)-0.18065(0)-0.0068893(35)-0.118722(5) = **-1.7816%**

Putney RET = -2.64216-2.11046(1)+0.005735(1355)-0.18065(1)-0.0068893(32)-0.118722(2) = **0.3956%**

Bob's RET is -1.7816%, Putney's RET is 0.3956%, Putney has a higher expected return by 2.1772%

(b) Between Bob and Putney, who is expected to obtain higher returns if hired by AMBTPM, and by how much?

If AMBTPM hires them, the TEN value will be zero.

Bob RET (TEN = 0) = **-1.188%**

Putney RET (TEN=0) = **0.6331%**

Putney is expected to obtain a higher return by 1.8211%

Question 3

(a) Can you prove at the 5 percent significance level that if Bob had attended Princeton instead of Ohio State, then the return of his current fund would be greater?

Since the question is directional (greater than), a one-tailed test on the SAT is conducted to test the hypothesis. The SAT coefficient is positive (0.005735) with t = 2.1563 and a two-tailed p-value =3.151%. For a one-tailed hypothesis, the one-tailed p-value is 1.5755% (=3.151%/2), which is below the 5 percent significance level. In this case, the null hypothesis is rejected, so higher SAT scores (Princeton) are associated with higher excess returns, in other words, at the 5 percent significance level, Bob would have a higher return if he attended Princeton.

(b) Can you prove at the 10 percent level of significance that if Bob were managing a growth fund instead of a growth and income fund, then he would achieve at least 1 percent higher average returns?

A growth and income fund (GRI=1) and a growth fund (GRI=0), and the coefficient for GRI is -2.11046, which means that the RET of the growth and income fund is 2.11046% lower than the RET of the growth fund. The t-statistic: $t = \frac{\widehat{\beta}- \beta}{SE} = \frac{-2.11046\%-(-1)}{0.738858} = -1.5029$

Since the 10 % one-tailed critical value is -1.28, we reject the null hypothesis (|-1.5|>|-1.28|). Therefore, Bob would achieve at least 1 % higher average return if he managed a growth fund.

Question 4

(a) Does the regression in Table 1 provide strong evidence for the claim that fund managers with MBAs perform worse than managers without MBAs? What is being held constant in this comparison? Discuss.

No, there is no strong evidence that managers with an MBA perform worse than those without one. The t-score (-0.2387) is near zero, the p-value (81.129%) is very large, indicating that the effect is not statistically significant at any conventional level. As a result, it fails to reject the null hypothesis that the MBA effect is zero. In this comparison, GRI, SAT, AGE, and TEN are all held constant.

(b) It has been suggested that fund managers without MBAs get higher expected returns because they invest in riskier stocks. If this were true, what effect would including an independent variable, Beta (with higher values corresponding to higher levels of systematic risk in the fund's portfolio), have on the coefficient of MBA in the regression of Table 1?

If non-MBA managers earn higher returns because they take more risk, then the MBA coefficient in Table 1 is capturing the effect of omitted risk. Adding Beta to the regression would control for systematic risk, so the higher returns due to risk would be attributed to Beta rather than to the MBA variable. As a result, the MBA coefficient would increase (become less harmful).

Question 5

(a) What is the lowest level of significance at which you can prove that the manager's age has a negative impact on his or her fund's performance holding the type of the fund, the manager's education, and years of experience at the fund constant?

The lowest level of significance at which we can conclude that manager age has a negative impact on fund performance is approximately 5 percent. The AGE coefficient is negative (−0.068893) with a t-statistic of −1.6474 and a two-tailed p-value of 10.006%. Because the hypothesis is directional, the relevant one-tailed p-value is about 5.003%, indicating that the null hypothesis can be rejected at the 5 percent level.

(b) A survivorship bias is thought to be present in analyzing fund manager performance in which a younger manager's survival in the industry is more closely linked to his/her performance than an older manager's survival. In other words, if a new manager does not perform successfully, he or she is not tolerated in the industry for long, but a more experienced manager may be forgiven a year or two of poor performance. Would the presence of this survivorship bias dampen or exacerbate the effect seen in Part (a)?

It will exacerbate the AGE effect mentioned in part (a). Poor-performing young managers are less likely to survive in the industry compared to older managers. The young managers who remain in the sample are intentionally selected for good performance, while older managers could stay for longer, even though they have the same poor performance. This selection makes younger managers look better relative to older managers than they would in an unbiased sample, which strengthens the observed negative relationship between AGE and performance.

Question 6

(a) "Streamline" the regression given in Table 1, that is, eliminate all variables that are not significant at the 15 percent level. Write down the new regression equation and check whether the specification satisfies the assumptions of linearity and homoskedasticity.

Eliminating all variables that are not significant at the 15 percent level means retaining only those with a p-value less than or equal to 15%, so that we will drop MBA (p=81.139%) and TEN (p=15.567%).

The new regression equation would be RET= -2.5839-2.1110(GRI)+0.00624(SAT)-0.09595(AGE)

After dropping the MBA and TEN data, the auxiliary regression of squared residuals on the explanatory variables yields an F-statistic with a p-value of 1 (Appendix 6a), and the residual plots (Appendix 6b) show no systematic pattern and a roughly constant spread. Hence, the streamlined model satisfies the assumptions of linearity and homoskedasticity.

(b) Compare the coefficient of AGE in the new and the old regressions. What can explain the sign (direction) of the change in this estimator? Discuss.

In the old regression, AGE is positively correlated with TEN because older managers tend to have longer tenure. When removing TEN from the regression model, part of its influence is shifted onto AGE, making the AGE coefficient more negative and reflecting both the direct effect of age and the omitted variables.

Question 7

(a) Run a regression to compare the average returns of growth and growth and income funds. Which type of fund yields a greater average return? Discuss whether or not the assumption of homoskedasticity is satisfied in this specification. What are the implications of this finding regarding the estimated difference in the average returns and the significance of the difference?

After running the regression (Appendix 7a), the intercept coefficient, average excess return of growth funds, is 0.396, and the GRI coefficient, growth and income funds, is -2.312, with a p-value of 0.00195, indicating that growth and income funds earn 2.312% less than growth funds. Hence, growth funds yield a greater average return. According to Appendix 7b, the homoskedasticity of the significance f-value 0.117 is not satisfied in the auxiliary regression, so the homoskedasticity assumption is likely violated.

While the estimated difference in average returns remains unbiased, unequal variance implies that the standard errors and significance tests may not be fully accurate.

(b) Redo the analysis in part a using Excel's TTEST function instead of a regression. Using this technique, can you prove at the 5 percent level of significance that the average returns of growth and growth and income funds differ?

By using Excel's TTEST function with a two-tailed test and unequal variance, the p-value is 0.00136, which is below the 5 percent level of significance, so the null hypothesis of equal average returns is rejected, meaning that the average returns of growth and growth and income funds are different (Appendix 7c).

Question 8

(a) You receive the prospectus of a growth fund started in the current year by a Princeton alum. What is the estimated RET (excess return relative to the return of the benchmark market portfolio) for this fund?

Since the MBA and AGE information are not given, we will use the sample means: MBA = 0.6204 and AGE = 43.93.

RET= -2.64216-2.11046(0)+0.005735(1355)-0.18065(0.6204)-0.0068893(43.93)-0.118722(0) = **1.9902%**

The estimated RET for this fund will be 1.9902%.

(b) Are you confident that this fund will "beat the market", that is, provide a return in excess of that of the benchmark market portfolio? Which standard error do we have to use in order to answer this question?

Normally, "beat the market" means RET>0. But, to determine whether a single fund will beat the market with confidence, a prediction interval is necessary to ensure the confidence of a single return. To capture the overall variability in fund returns, we don't use the standard error of an individual coefficient, instead, I use the standard error of the regression shown in Table 1. In Table 1, the standard error of regression is 8.354658, and the 95% critical value is 1.9644.

A 95% prediction level: $= 1.9902\% \pm 1.9644 \times 8.3547$ = [-14.4235%, 18.4039%]

Within this interval, it includes 0, so even though the RET is positive, we cannot conclude that it will outperform the market.

(c) Suppose that you manage to identify a large number of growth funds started recently by Princeton graduates. By investing equally in all of these funds, how likely is it that your return will exceed that of the benchmark market portfolio by more than 1.5 percent? Which standard error did you use in your answer?

Unlike the previous question, we now invest equally in a large number of growth funds, so the portfolio return represents an average across funds, which reduces random variation. As a result, the portfolio's return is expected to be closer to the estimated excess return of 1.9902%.

To assess whether this average excess return exceeds the benchmark by more than 1.5%, the relevant uncertainty is measured by the standard error of the mean return, rather than the prediction error for a single fund. Although the expected excess return exceeds 1.5%, it does so by only 0.4902%. Given the remaining uncertainty around the estimated mean, we cannot be highly confident that the portfolio will outperform the benchmark market portfolio by more than 1.5%.

Question 9

Suppose that you gain access to a much larger sample of random observations of the same variables that you have in the current dataset. Do you expect that any of your answers to Parts (a)-(c) of Question 8 will change, and if so, how? Discuss.

If we have a much larger sample dataset, the answer will not change a lot. For part (a), the estimated RET for a Princeton growth fund is one prediction based on the regression model. Having a larger sample size might shift a little bit due to sampling variation, but the overall result won't change that much. In addition, a larger sample would reduce the standard errors of the coefficients, leading to more precise estimates. For part (b), using a larger sample will reduce uncertainty, but the return of a single fund would remain highly volatile, so the conclusion will not change even with more data. For part(c), the conclusion will become stronger since those estimates will be more accurate with more observations.

Question 10

(a) Based on the dataset, can you prove at the 5 percent level of significance that among fund managers with the same educational background and same experience with the same fund, those managing growth and income funds are, on average, older?

After running the regression, the GRI coefficient is positive (1.4242), and the two-tailed p-value is 0.062, so the one-tailed p-value is 0.031. Since the one-tailed p-value is less than 0.05, meaning the null hypothesis that growth and income fund managers are not older is rejected, in other words, growth and income fund managers are, on average, older, holding other factors constant.

(b) Using the regression developed in part a, provide an 80 percent confidence interval for the average age difference between managers who graduated from the same college in the United States and have managed a growth fund for the same number of years, but differ in whether or not they have an MBA. Are the (otherwise comparable) managers with MBAs younger or older, on average? Discuss (conjecture) why this is the case.

According to Appendix 10, the coefficient on MBA is -1.8793, and the standard error is 0.7780. To calculate the 80% confidence level, the critical value is 1.283.

An 80% confidence level: $-1.8793\% \pm 1.283 \times 0.778 = [-2.8773, -0.8813]$

Because the interval is negative, it indicates that MBA managers are, on average, younger. This might be because an MBA is more common among younger managers when they enter the industry, and it might not be necessary for older managers, who are more experienced.

Question 11

Based on your analysis of the case, which candidate do you support for AMBTPM's job opening: Bob or Putney? Discuss.

In my opinion, I support hiring Putney for this position. According to Table 1, she has a higher excess return (0.3956%) than Bob in their current funds, and a higher RET (0.6331%) if both are hired by AMBTPM, indicating she will outperform Bob by 1.8211%. Moreover, no evidence that obtaining an MBA will negatively affect fund performance, so Putney does not have a disadvantage in this component. In addition, the SAT is positively correlated to the excess return, so her educational background is expected to perform better than Bob's while holding other factors constant. By contrast, Bob is less suitable because he has lower current and predicted excess returns than Putney, lacks educational characteristics that positively affect performance, and does not benefit from any compensating factors in the regression. Overall, based on the analysis, Putney is expected to produce a higher excess return and is more suitable for this position.

Appendices

Appendix 1. Auxiliary Regression on Squared R

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.181046226
R Square	0.032777736
Adjusted R Squa	0.023721348
Standard Error	151.2680227
Observations	540

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	414083.7495	82816.74989	3.619294498	0.003145206
Residual	534	12218995.85	22882.0147		
Total	539	12633079.6			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	141.977901	60.59172222	2.343189726	0.01948567	22.95053091	261.0052711	22.95053091	261.0052711
GRI	-24.11473379	13.37763656	-1.802615409	0.072012332	-50.39398183	2.164514248	-50.39398183	2.164514248
SAT	-0.064517073	0.04815368	-1.339816043	0.180875023	-0.15911095	0.030076804	-0.15911095	0.030076804
AGE	0.66044674	0.757146003	0.872284524	0.383445209	-0.82690325	2.147796731	-0.82690325	2.147796731
MBA	-37.78167082	13.69966381	-2.757853868	0.006017539	-64.69351446	-10.86982718	-64.69351446	-10.86982718
TEN	1.221883408	1.511875515	0.808190486	0.419340818	-1.748069581	4.191836397	-1.748069581	4.191836397

Appendix 6a. Auxiliary Regression on Squared R without MBA and TEN

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	2.4153E-08
R Square	5.8336E-16
Adjusted R Sc	-0.005597
Standard Err	16.7102948
Observations	540

ANOVA

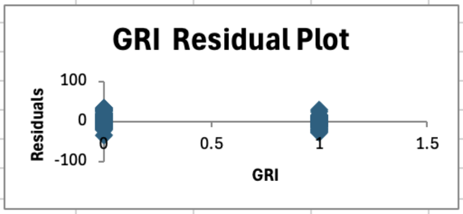
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	8.7311E-11	2.9104E-11	1.0423E-13	1
Residual	536	149669.398	279.233952		
Total	539	149669.398			

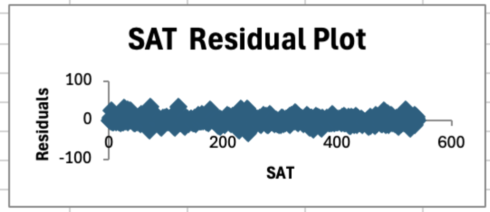
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	3.37E-15	6.68070061	5.0443E-16	1	-13.123566	13.1235663	-13.123566	13.1235663
GRI	-4.397E-15	1.47715974	-2.976E-15	1	-2.9017322	2.90173215	-2.9017322	2.90173215
SAT	-1.504E-18	0.00518631	-2.9E-16	1	-0.010188	0.01018798	-0.010188	0.01018798
AGE	4.203E-17	0.07311022	5.7489E-16	1	-0.1436177	0.1436177	-0.1436177	0.1436177

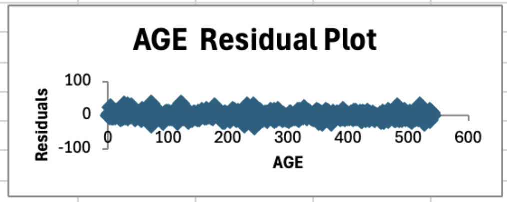
Appendix 6b. Regression without MBA and TEN

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.1991312
R Square	0.03965323
Adjusted R Sq	0.03427816
Standard Err	8.35514739
Observations	540







ANOVA

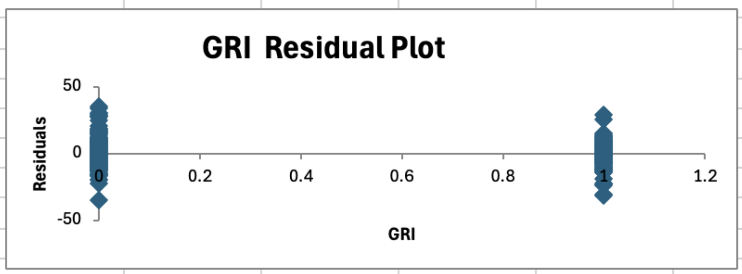
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	1544.98248	514.994158	7.37724271	7.5053E-05
Residual	536	37417.3495	69.8084879		
Total	539	38962.332			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-2.5839223	3.34035031	-0.7735483	0.43953898	-9.1457054	3.97786087	-9.1457054	3.97786087
GRI	-2.1110054	0.73857987	-2.8581952	0.00442628	-3.5618715	-0.6601393	-3.5618715	-0.6601393
SAT	0.00624165	0.00259315	2.40697141	0.01642256	0.00114766	0.01133564	0.00114766	0.01133564
AGE	-0.0959589	0.03655511	-2.6250465	0.00891103	-0.1677677	-0.02415	-0.1677677	-0.02415

Appendix 7a. Regression on Growth and Income Fund and Growth Fund

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.133016057
R Square	0.017693272
Adjusted R Squ	0.015867423
Standard Error	8.434413582
Observations	540



ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	689.3711188	689.3711188	9.690435583	0.001950772
Residual	538	38272.96087	71.13933247		
Total	539	38962.33199			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.395923653	0.466424137	0.84884898	0.396342862	-0.52031208	1.312159381	-0.52031208	1.312159381
GRI	-2.31185049	0.742656685	-3.11294645	0.001950772	-3.77071279	-0.85298819	-3.77071279	-0.85298819

Appendix 7b. Auxiliary Regression on Squared R (GRI)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.06754075
R Square	0.00456175
Adjusted R Sc	0.0027115
Standard Err	160.344174
Observations	540

ANOVA

	df	SS	MS	F	Significance F
Regression	1	63387.8529	63387.8529	2.46546973	0.11696082
Residual	538	13832116.6	25710.254		
Total	539	13895504.5			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	79.6200934	8.86705306	8.97931848	4.5651E-18	62.2018035	97.0383833	62.2018035	97.0383833
GRI	-22.168496	14.1184294	-1.5701814	0.11696082	-49.902501	5.56550951	-49.902501	5.56550951

Appendix 7c. TTEST on Growth and Income fund and Growth fund

F3 =TTEST(C2:C541,D2:D541,2,3)

	A	B	C	D	E	F
1	RET	GRI	ret_growth	ret_growthincome		
2	-0.07574	1		-0.0757434		TTEST
3	5.53836	0	5.538359			0.00136148
4	-5.59826	1		-5.598259		
5	25.2952	0	25.29515			
6	-4.28823	0	-4.288226			
7	-1.9606	1		-1.960604		
8	-3.25932	1		-3.259319		
9	0.00888	0	0.0088848			
10	-0.715	0	-0.7150039			
11	11.9162	0	11.91617			
12	-1.21176	1		-1.211762		
13	-4.04327	0	-4.043267			
14	9.99204	0	9.992036			
15	3.19305	1		3.193048		
16	1.05778	0	1.057781			
17	-0.87617	1		-0.8761674		
18	-3.91513	0	-3.915133			
19	-0.34368	1		-0.3436834		
20	1.10251	0	1.102513			
21	-7.59635	0	-7.596345			
22	0.10617	0	0.1061745			
23	-6.04005	1		-6.040053		
24	-4.86411	0	-4.864105			
25	-1.50708	0	-1.50708			
26	-1.37139	0	-1.371387			
27	30.9879	0	30.98786			
28	0.79811	1		0.7981062		
29	-3.85701	1		-3.857007		
30	17.9059	0	17.90589			
31	4.64966	1		4.64966		

Appendix 10 Regression of Fund Manager Age on Other Factors

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.492431979
R Square	0.242489254
Adjusted R Square	0.236825622
Standard Error	8.637550614
Observations	540

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	12777.28821	3194.322052	42.81515191	3.59058E-31
Residual	535	39914.89512	74.6072806		
Total	539	52692.18333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	32.18608414	3.167679053	10.16077816	2.69268E-22	25.96347002	38.40869826	25.96347002	38.40869826
GRI	1.424150325	0.761390514	1.870459769	0.061965762	-0.071531303	2.919831953	-0.071531303	2.919831953
SAT	0.007759397	0.002729081	2.84322733	0.004636614	0.002398369	0.013120425	0.002398369	0.013120425
MBA	-1.879255026	0.778033392	-2.415391223	0.016052461	-3.407630056	-0.350879995	-3.407630056	-0.350879995
TEN	0.942056623	0.076118123	12.37624617	4.0909E-31	0.792529573	1.091583673	0.792529573	1.091583673